

A nice placeholder with no name

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Abstract

In [P1110][3] Jeffrey Yesskin and JF Bastien explore the design space of placeholders with the goal of not naming entities for which a name would provide no additional information. In this paper, we propose a concrete solution for the use of `_` (U+005F `_` LOW LINE) as such placeholder for variable declarations and pattern matching **in a fully backward compatible manner**.

Tony tables

Before	After
<pre>std::lock_guard namingIsHard(mutex); // Structured binding [[maybe_unused]] auto [x, y, iDontCare] = f(); // Pattern matching inspect(foo) { __ => bar; };</pre>	<pre>std::lock_guard _(mutex); // Structured binding auto [x, y, _] = f(); // Pattern matching inspect(foo) { _ => bar; };</pre>

Revisions

Revision 4

- Wording improvements

Revision 3

- Add wording
- R2 mentioned that we could allow `_` for global variables at namespace scope in module implementation units. While true, this seems too much of a weird allowance.

Revision 2

- A few more details about interactions with P2011

Revision 1

- Specify that placeholders can appear as class members
- Make use of `_` ill-formed once a placeholder has been declared in the same scope

Motivation

Both [P1110][3], [P1469][2] and [P0577][4] go over the motivation both for a placeholder syntax in general and the single underscore `_`:

- Some variable, like locks, `scope_guard` and other RAII objects are only used for their side effects.

```
std::lock_guard lock(mutex);
```

- Some structured bindings may not be used, and we would like a syntax to specify that this is the case. Introducing names for variables that are not used to convey less meaning than `_`, which convey the "I don't care meaning". It is furthermore not possible to apply attributes to structured bindings and as such, it is not possible to carry the developer's intent to the compiler. The current wording advises:

Recommended practice: For an entity marked `[[maybe_unused]]`, implementations should not emit a warning that the entity or its structured bindings (if any) are used or unused. For a structured binding declaration not marked `[[maybe_unused]]`, implementations should not emit such a warning unless all of its structured bindings are unused.

- We need a wildcard token for pattern matching. The proposed `_` is, as explained by [P1469][2] an industry wide-standard and we believe having a single token `_` for both wildcards and placeholders contribute to make C++ more consistent with itself and with other languages (making easier to teach, etc)

Design considerations

The design as presented in this paper takes the following design parameters into consideration

- `_` is recognized as some kind of placeholder in many existing languages, including `go`, `python`, `rust`, `C#`.
- `_` is used in existing code and popular frameworks.
- `_` might be defined as a macro function in some files depending on the `gettext` library.

- `_` already carries the meaning of "I don't want to use this variable". Variables that developers care about have more meaningful names.
- `_` unnamed variables cannot have linkage.
- `_` and `__` are similar enough that we should not give meaning to both.
- Tokens are rare, using a different token for placeholder may needlessly restrict the design space for more important features.
- For consistency, we would like to use the same token as the pattern matching placeholder.

Proposal

We propose that when `_` is used as the identifier for the declaration of a variable, non static class member variable, lambda capture or structured binding. the introduced name is implicitly given the `[[maybe_unused]]` attribute.

[Example:

```
auto _ = foo(); // equivalent to [[maybe_unused]] auto _ = f();
```

— end example]

When a variable with automatic storage duration, a non-static class member variable, structured binding or lambda capture is introduced by the identifier `_` it can redefine an existing declaration, `[basic.scope.declarative]` in the same scope. After redeclaration, if the variable is used, the program is ill-formed.

[Example:

```
namespace a {
    auto _ = f(); // Ok, declare a variable "_"
    auto _ = f(); // error: "_" is already defined in this namespace scope
}
void f() {
    auto _ = 42; // Ok, declare a variable "_"
    auto _ = 0; // Ok, re-declare a variable "_"
    {
        auto _ = 1; // Ok, shadowing
        assert( _ == 1 ); // Ok
    }
    assert( _ == 42 ); // ill-formed: Use of a redeclared placeholder variables
}
```

— end example]

In contexts where the grammar expects a pattern matching pattern, `_` represents the wildcard pattern.

```
inspect(i) {
    0 => 0;
    _ => 1; // wildcard pattern
};
```

NTTP

We do not propose to support

```
template <auto _, auto _>  
void f();
```

This is to avoid adding unnecessary complexity - template arguments can already be left unnamed without loss of functionality, and declarations in a templated entity are not allowed to have the same name as that of a template parameter currently, which would introduce additional complexities.

requires clause parameter names and parameter names

In Varna, EWG decided not to support function parameter names and requires clauses parameter as these can remain unnamed without semantics differences.

Both EWG and CWG expressed a desire of consistency between NTTP, function parameters and require clauses parameters, and not supporting either introduce less complexity while offering a simple model:

Any variable declaration that *has to* be named, can be named with a `_` placeholder variable. Any variable declaration that *does not have to* be named can just remain unnamed, with the same effect.

Should declaring variables called `_` be deprecated?

This paper does not propose to deprecate variables called `_`, as a few frameworks have legitimate and widely deployed usages of it (see the section about google mock), and a depreciation is not necessary for the well-behaving of this proposal. We do however think that such depreciation may be useful to consider in a longer time frame.

Alternative Design

A previous version of this proposal suggested that `_` should refer to the first declaration in scope, but many preferred the approach in this current revision.

Pattern matching

Citing [P1371R0][1], `_` can be used as a pattern matching wildcard without ambiguities

Even though `_` is a valid identifier, it does not introduce a name as doing so would result in redeclaration errors in the case where multiple wildcard `_` identifiers are used. It is possible for users to have already introduced `_` as a type or variable name in the same scope where an `inspect` statement is used. As a highly-visible example, the authors are aware of the use of `_` as a name in the popular “Google Mock” library. Idiomatically this is accessed

by introducing `_` into the current namespace or block scope with a `using` declaration. Using the wildcard pattern in cases like this is unambiguous since the expression pattern requires a `^` introducer for primary expressions. `_` will always match against an existing name and `_` will always represent the wildcard pattern. An existing `_` name can be used without ambiguity in the matched statement to which control is passed. Naturally, the impact of defining `_` in the pre-processor cannot be predicted or controlled by this paper and is thus liable to result in an ill-formed program.

Impact on P2011 and other pipeline rewrite musing

Placeholders in pipeline rewrite have a “Bind to this” semantic rather than a “I don’t care” semantic. As such, these proposals could use a different syntax. However, there are no syntactic ambiguities between the two features, P2169 impacts only declarations, not expressions.

Independently of P2169, P2011 makes the following ambiguous

```
int _ = 0; // Not a placeholder
f() |> g(_, _); // placeholder or variable ?
```

This is simply resolved by having `_` always be a placeholder in pipeline expression, and any `_` variable can be access either through a reference or by wrapping it in parentheses.

```
int _ = 0; // Not a placeholder
auto & placeholder = _;
f() |> g(_, placeholder); // OK
f() |> g(_, (_)); // OK
```

And so,

- There are no adverse synergies between this proposal and pipeline rewrite
- There are no adverse synergies between this proposal and pattern matching
- This proposal and pattern matching use the `_` token for similar semantics (“don’t care”), whether the pipeline rewrite has a different semantic (“bind to this argument”). However, both are placeholders and an argument can also be made to use the same syntax in both context, it really boils down to what we think is less surprising.

Impact on existing code

Google Mock

Google Mock appear to constitute the most common use of the `identifier`. With this proposal, Google Mock continues to work. Variables named `_` may shadow the `testing::_` global variable introduced by google mock:

```
namespace testing {
    const internal::AnythingMatcher _ = {};
```

```
}
```

Because the proposal limits the use of placeholder to function scope (because of linker considerations), there is little risk that increased use of `_` causes shadowing issues for users of this framework.

It is possible to use the feature proposed in this paper along with google mock as long as the using namespace `testing`; appears before any declaration of `_` in that code.

[Compiler explorer link](#)

Gettext

Some projects using Gettext define `_` as a function-like macro.

```
# define _(msgid) gettext (msgid)
```

It is important to know that this is not provided in any gettext header. It is also technically undefined as the `_` is reserved at global scope. But, regardless, defining this macro would not prevent the use of the proposed syntax, as `_` is only expanded by the preprocessor in this case if followed by parentheses:

```
constexpr const char* gettext(int) { return nullptr;}
#define _(msgid) gettext (msgid)

int main() {
    constexpr auto _ = _(42);
    auto _ = 42;
    static_assert(_ == nullptr);
}
```

[Compiler explorer link](#)

Implementation

R0 of this proposal has been implemented in Clang and the implementation is available on [Compiler Explorer](#).

The implementation consists of ignoring existing `_` variables in the same scope when declaring placeholder variables. Note that name lookup is not affected by this proposal, some variables are simply skipped over based on their name. The variables otherwise conserve their name for diagnostics purposes.

The current version of this proposal can be emulated with the `-Werror` flag [Compiler Explorer](#).

Wording



General

[basic.scope.scope]

[Editor's note: Modify [basic.scope.scope]/p5]

A declaration is *name-independent* if its name is `_` and it declares a variable with automatic storage duration, a structured binding not inhabiting a namespace scope, the variable introduced by an *init-capture*, or a non-static data member.

Recommended practice: Implementations should not emit a warning that a name-independent declaration is used or unused.

Two declarations *potentially conflict* if they correspond and cause their shared name to denote different entities[basic.link]. The program is ill-formed if, in any scope, a name is bound to two declarations A and B that potentially conflict and **one** A precedes **the other** B [basic.lookup], unless B is name-independent.

[*Note:* An *id-expression* that names a unique name-independent declaration is usable until an additional declaration of the same name is introduced in the same scope [basic.lookup.general]. — *end note*]

[*Note:* Overload resolution can consider potentially conflicting declarations found in multiple scopes (e.g. via *using-directive* s or for operator functions), in which case it is often ambiguous. — *end note*] [*Example:*

```
void f() {
    int x,y;
    void x();           // error: different entity for x
    int y;              // error: redefinition
}
enum { f };            // error: different entity for ::f
namespace A {}
namespace B = A;
namespace B = A;        // OK, no effect
namespace B = B;        // OK, no effect
namespace A = B;        // OK, no effect
namespace B {}          // error: different entity for B

void g() {
    int _;
    _ = 0;              // OK
    int _;              // OK, name independent declaration
    _ = 0;              // error: two non-function declarations in the lookup set
}

void h () {
    int _;              // #1
    _ ++;               // OK
    static int _;       // error: conflicts with #1 because static variables are not
                        // name-independent
}
```

— *end example*]

[...]

❖ Block scope

[basic.scope.block]

[...]

If a declaration [that is not a name-independent declaration and](#) whose target scope is the block scope S of a

- *compound-statement* of a *lambda-expression*, *function-body*, or *function-try-block*,
- substatement of a selection or iteration statement that is not itself a selection or iteration statement, or
- *handler* of a *function-try-block*

potentially conflicts with a declaration whose target scope is the parent scope of S , the program is ill-formed. [*Example*:

```
if (int x = f()) {  
    int x;           // error: redeclaration of x  
}  
else {  
    int x;           // error: redeclaration of x  
}
```

— *end example*]

[...]

❖ Program and linkage

[basic.link]

[Editor's note: Modify [basic.link]/p8 as follow]

Two declarations of entities declare the same entity if, considering declarations of unnamed types to introduce their names for linkage purposes, if any[dcl.typedef,dcl.enum], they correspond[basic.scope.scope], have the same target scope that is not a function or template parameter scope, [neither is a name-independent declaration](#), and either

- they appear in the same translation unit, or
- they both declare names with module linkage and are attached to the same module, or
- they both declare names with external linkage.

[*Note*: There are other circumstances in which declarations declare the same entity[dcl.link,temp.type,temp.spacial]. — *end note*]

[...]

❖ The using declaration

[namespace.udecl]

Modify [namespace.udecl]/p10

If a declaration named by a *using-declaration* that inhabits the target scope of another declaration **B** potentially conflicts with it [basic.scope.scope] and either is reachable from the other, the program is ill-formed unless B is name-independent and the *using-declaration* precedes B.

[Example:

```
int _;
void f() {
    int _;    // B
    _ = 0;
    using ::_; // error: using-declaration does not precede B
}
```

— end example]

If two declarations named by *using-declarations* that inhabit the same scope potentially conflict, either is reachable from the other, and they do not both declare functions or function templates, the program is ill-formed. [Note: Overload resolution possibly cannot distinguish between conflicting function declarations. — end note]

Feature test macros

[Editor's note: Add a new macro in [tab:cpp.predefined.ft] : `__cpp_placeholder_variables` set to the date of adoption].

```
#define __cpp_placeholder_variables 2023XX // date of adoption
```

Rebuttal some wilder ideas explored by P1110

[P1110] explores the different places where a placeholder may be used.

Enums

The code on the left (P1110) can be written in standard C++ in a way that expresses the intent as clearly, if not more.

```
enum MmapBits {
    Shared,
    Private,
    ...,
    Fixed,
    Rename,
    //...
};
```

```
enum MmapBits {
    Shared,
    Private,
    Fixed = Private + 3,
    Rename,
    //...
};
```

Concept-constrained declarations

The following example from P1110 is a historical artifact as C++20 uniformizes `auto` for this purpose.

```
template<Integral __ N> class integral_constant { ... };
Numeric multiplyAdd(Numeric __ x, Numeric __ y, Numeric __ z) {
    Numeric __ multiplied = x * y;
    return multiplied + z;
}
```

Here is the C++20 equivalent

```
template<Integral auto N> class integral_constant { ... };
Numeric multiplyAdd(Numeric auto x, Numeric auto y, Numeric auto z) {
    Numeric auto multiplied = x * y;
    return multiplied + z;
}
```

Other explored areas, including placeholders for function names, type declaration, using and concepts are not useful, either because C++ does already provide the desired facilities such as anonymous type, or because the entity needs a name (concepts, functions and using declarations without a name cannot be used)

Acknowledgments

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References

- [1] Sergei Murzin, Michael Park, David Sankel, and Dan Sarginson. P1371R0: Pattern matching. <https://wg21.link/p1371r0>, 1 2019.
- [2] Sergei Murzin, Michael Park, David Sankel, and Dan Sarginson. P1469R0: Disallow `_` usage in c++20 for pattern matching in c++23. <https://wg21.link/p1469r0>, 1 2019.
- [3] Jeffrey Yasskin and JF Bastien. P1110R0: A placeholder with no name. <https://wg21.link/p1110r0>, 6 2018.
- [4] Zhihao Yuan. P0577R0: Keep that temporary! <https://wg21.link/p0577r0>, 2 2017.
- [N4861] Richard Smith *Working Draft, Standard for Programming Language C++* <https://wg21.link/N4861>